Examiner only

Answer **all** *questions.*

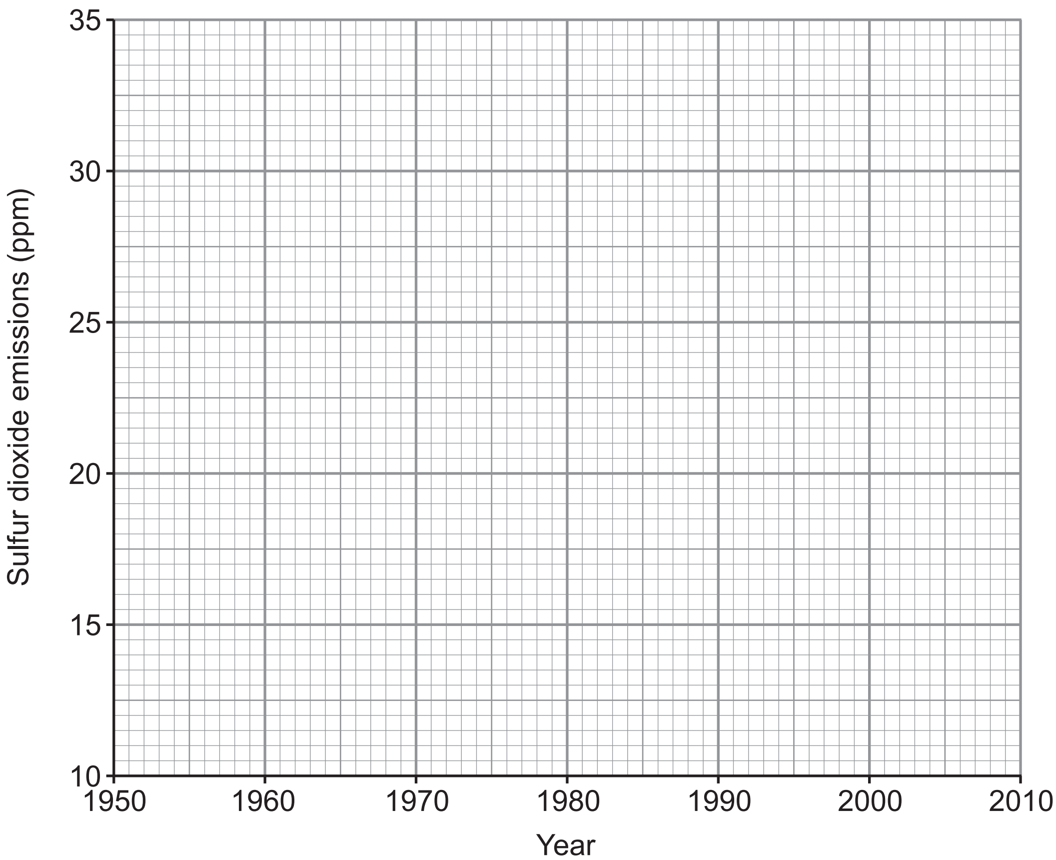
1. Burning fossil fuels containing sulfur causes sulfur dioxide, SO_2, to be released into the atmosphere.

The table shows sulfur dioxide emissions in the UK between 1950 and 2010.

Year	Sulfur dioxide emissions (ppm)
1950	12.0
1960	16.0
1970	21.5
1980	29.5
1990	29.0
2000	24.0
2010	18.5

ppm = parts per million

(a) (i) On the grid plot the sulfur dioxide emissions against the year and draw a suitable line. [3]



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(ii) Describe how sulfur dioxide emissions changed between 1950 and 2010. [2]

....

....

....

(iii) The UK government introduced a regulation to reduce sulfur dioxide emissions in the 1980s. From your graph, state why it is difficult to decide exactly the year when the regulation came into force. [1]

....

....

....

(b) Sulfur dioxide can be converted to sulfur and water by reacting it with hydrogen sulfide, H_2S.

Complete and balance the symbol equation for this reaction. [2]

$$SO_2 + \boxed{}\,H_2S \longrightarrow \boxed{}\,S + \boxed{}\ \ldots\ldots\ldots\ldots$$

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(iii) Put a tick (✓) in the box next to the **two** statements which describe the conclusions that can be drawn from the data in **Figure 1**. [2]

lead contamination in road-side soil decreases in towns and in the countryside as the distance from the centre of the road increases ☐

the decrease in lead contamination between 12 m and 42 m from the centre of the road is greater in towns than in the countryside ☐

lead contamination in road-side soil at all distances is much greater in towns than in the countryside ☐

lead contamination in road-side soil decreases between 12 m and 42 m from the centre of the road in the countryside ☐

there is approximately 50 % more lead contamination in road-side soil in towns compared to the countryside between 12 m and 28 m from the centre of the road ☐

there is an overall decrease of nearly 70 % in the lead contamination of road-side soil in towns between 12 m and 42 m from the centre of the road ☐

(iv) In the mid-1970s the use of lead in paints and the manufacture of cars that used leaded petrol was banned. Put a tick (✓) in the box next to the statement which **best** describes what happened after the ban. [1]

paint and petrol were lead-free by the mid-1970s anyway ☐

paint and petrol were not lead-free until 1980 ☐

the use of lead-based paint and leaded petrol increased until 1980 before decreasing ☐

it took 15 years for paint and petrol to become lead-free ☐

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(b) (i) Lead can form a number of oxides with different formulae. Red lead, Pb_3O_4, is used to make batteries. It is manufactured by reacting lead carbonate with oxygen. Carbon dioxide is also produced.

Balance the equation for this reaction. [1]

$$\square\, PbCO_3 + O_2 \longrightarrow \square\, Pb_3O_4 + \square\, CO_2$$

(ii) 11.36 g of a lead oxide contains 10.18 g of lead. Calculate the empirical formula of this oxide. [3]

$A_r(Pb) = 207$ $A_r(O) = 16$

Empirical formula ....

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END OF PAPER

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FORMULAE FOR SOME COMMON IONS

POSITIVE IONS		NEGATIVE IONS	
Name	**Formula**	**Name**	**Formula**
aluminium	Al^{3+}	bromide	Br^{-}
ammonium	NH_4^{+}	carbonate	CO_3^{2-}
barium	Ba^{2+}	chloride	Cl^{-}
calcium	Ca^{2+}	fluoride	F^{-}
copper(II)	Cu^{2+}	hydroxide	OH^{-}
hydrogen	H^{+}	iodide	I^{-}
iron(II)	Fe^{2+}	nitrate	NO_3^{-}
iron(III)	Fe^{3+}	oxide	O^{2-}
lithium	Li^{+}	sulfate	SO_4^{2-}
magnesium	Mg^{2+}		
nickel	Ni^{2+}		
potassium	K^{+}		
silver	Ag^{+}		
sodium	Na^{+}		
zinc	Zn^{2+}		

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THE PERIODIC TABLE

Group

1	2											3	4	5	6	7	0
						1 H Hydrogen 1											4 He Helium 2
7 Li Lithium 3	9 Be Beryllium 4											11 B Boron 5	12 C Carbon 6	14 N Nitrogen 7	16 O Oxygen 8	19 F Fluorine 9	20 Ne Neon 10
23 Na Sodium 11	24 Mg Magnesium 12											27 Al Aluminium 13	28 Si Silicon 14	31 P Phosphorus 15	32 S Sulfur 16	35.5 Cl Chlorine 17	40 Ar Argon 18
39 K Potassium 19	40 Ca Calcium 20	45 Sc Scandium 21	48 Ti Titanium 22	51 V Vanadium 23	52 Cr Chromium 24	55 Mn Manganese 25	56 Fe Iron 26	59 Co Cobalt 27	59 Ni Nickel 28	63.5 Cu Copper 29	65 Zn Zinc 30	70 Ga Gallium 31	73 Ge Germanium 32	75 As Arsenic 33	79 Se Selenium 34	80 Br Bromine 35	84 Kr Krypton 36
86 Rb Rubidium 37	88 Sr Strontium 38	89 Y Yttrium 39	91 Zr Zirconium 40	93 Nb Niobium 41	96 Mo Molybdenum 42	99 Tc Technetium 43	101 Ru Ruthenium 44	103 Rh Rhodium 45	106 Pd Palladium 46	108 Ag Silver 47	112 Cd Cadmium 48	115 In Indium 49	119 Sn Tin 50	122 Sb Antimony 51	128 Te Tellurium 52	127 I Iodine 53	131 Xe Xenon 54
133 Cs Caesium 55	137 Ba Barium 56	139 La Lanthanum 57	179 Hf Hafnium 72	181 Ta Tantalum 73	184 W Tungsten 74	186 Re Rhenium 75	190 Os Osmium 76	192 Ir Iridium 77	195 Pt Platinum 78	197 Au Gold 79	201 Hg Mercury 80	204 Tl Thallium 81	207 Pb Lead 82	209 Bi Bismuth 83	210 Po Polonium 84	210 At Astatine 85	222 Rn Radon 86
223 Fr Francium 87	226 Ra Radium 88	227 Ac Actinium 89															

Key

A_r	relative atomic mass
Symbol	
Name	
Z	atomic number

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